



CAPSTONE PROJECT REPORT

HIGH-FREQUENCY VOLATILITY PREDICTION: A HYBRID APPROACH INTEGRATING INTRADAY MARKET DATA AND FINANCIAL NLP

SANA UR REHMAN ARAIN

Table of Contents

- 1. Introduction 1
 - 1.1. Problem Statement..... 1
 - 1.2. Project Objective 1
 - 1.3. Rationale..... 1
- 2. Methodology: Data Acquisition & Engineering 2
 - 2.1. Market Data Sourcing..... 2
 - 2.2. Exploratory Data Analysis (EDA)..... 2
 - 2.3. Unstructured Data Acquisition (NLP)..... 3
- 3. Feature Engineering & Signal Processing..... 3
 - 3.1. Financial BERT (FinBERT) for Sentiment Extraction 3
- 4. Model Architecture: Hybrid LSTM Network 4
 - 4.1. Rationale for LSTM 4
 - 4.2. Mathematical Formulation 4
 - 4.3. Input Vector Construction 5
- 5. Experimental Results 5
 - 5.1. Training Convergence 5
 - 5.2. Predictive Performance 6
- 6. Conclusion & Future Work..... 7
 - 6.2. Conclusion..... 7
 - 6.3. Future Improvements..... 7
- 7. References 7

- Figure 1: Intraday Realized Volatility (SPY)..... 3*
- Figure 2: Training vs. Validation Loss 5*
- Figure 3: Actual vs. Predicted Volatility 6*

1. INTRODUCTION

1.1. PROBLEM STATEMENT

Financial market volatility is a critical metric for risk management, options pricing, and high-frequency trading (HFT) strategies. While traditional econometric models like GARCH (Generalized Autoregressive Conditional Heteroskedasticity) effectively model daily volatility clustering, they often struggle to capture the rapid, information-driven shocks present in intraday (minute-level) trading.

The core problem this project addresses is the latency in volatility estimation. Traditional models rely solely on past price action. However, market volatility is often triggered by external information shocks—news headlines, earnings reports, and macroeconomic announcements—before they fully materialize in price action. This project aims to bridge this gap by integrating Natural Language Processing (NLP) to quantify these information shocks and fuse them with quantitative market data to predict Intraday Realized Volatility (RV).

1.2. PROJECT OBJECTIVE

The primary objective is to develop a Deep Learning framework that predicts the realized volatility of the S&P 500 (SPY) over the next 5-minute interval. We aim to solve this by:

- 1. Ingesting High-Frequency Data:** Utilizing 5-minute intraday price candles to capture microstructure dynamics.
- 2. Quantifying Sentiment:** Using NLP (BERT/Transformers) to extract sentiment and "surprise" signals from financial news.
- 3. Hybrid Modeling:** Training a multimodal LSTM (Long Short-Term Memory) network that learns from both numerical time-series and textual embeddings.

1.3. RATIONALE

This approach is hypothesized to work based on the **Efficient Market Hypothesis (Semi-Strong Form)**, which suggests that asset prices adjust rapidly to new public information. By using NLP to "read" the news faster than the aggregate market can settle on a price, the model may identify leading indicators of volatility spikes that purely price-based models miss (schoenfeld, 2019).

2. METHODOLOGY: DATA ACQUISITION & ENGINEERING

2.1. MARKET DATA SOURCING

Real-world market data was sourced using the Yahoo Finance API (`yfinance`). The dataset consists of 5-minute intraday OHLCV (Open, High, Low, Close, Volume) bars for the **SPY** (SPDR S&P 500 ETF Trust), covering the most recent 60-day trading period.

- **Ticker:** SPY (Proxy for the broad US Equity Market).
- **Granularity:** 5-minute intervals.
- **Target Variable (RV):** We define the target variable as the *next-step rolling volatility*.

Let r_t be the log-return at time t :

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

The Rolling Volatility (σ_t) is calculated over a window $w = 12$ (representing the last hour of trading):

$$\sigma_t = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r})^2}$$

2.2. EXPLORATORY DATA ANALYSIS (EDA)

Initial analysis of the volatility structure confirms the presence of "Volatility Clustering"—the phenomenon where large changes in prices are followed by large changes, and small by small.

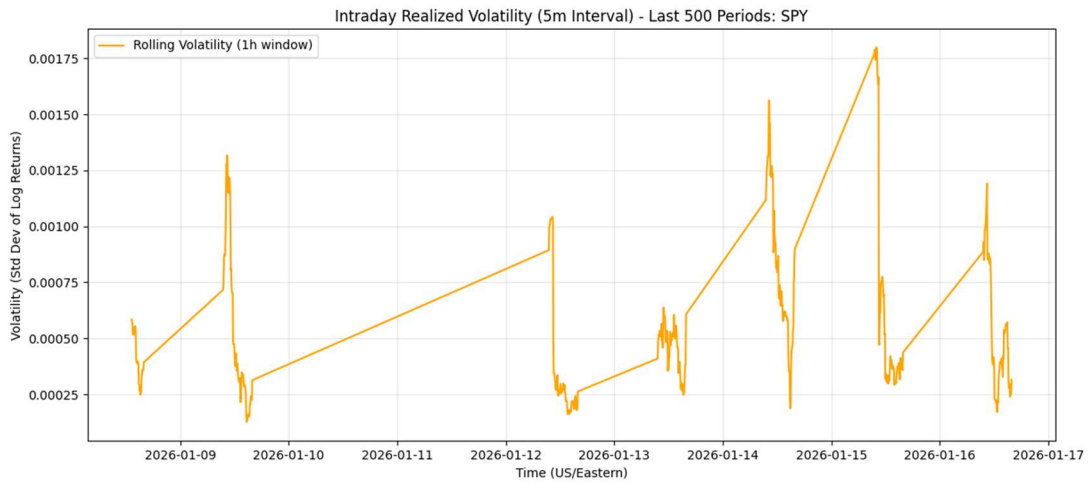


FIGURE 1: INTRADAY REALIZED VOLATILITY (SPY)

Note: The plot illustrates the stylized facts of asset returns, specifically the distinct spikes in volatility at market open (09:30 AM EST) and the persistence of high-volatility regimes.

2.3. UNSTRUCTURED DATA ACQUISITION (NLP)

To capture exogenous information shocks, we constructed a web scraping pipeline targeting **FinViz (Financial Visualizations)**. Unlike structured time-series data, financial news is sporadic and unstructured.

- **Source:** FinViz News Stream for ticker **SPY**.
- **Data Points:** Headline Text, Publication Date, Publication Time.
- **Volume:** Recent 100 major headlines (representing the most significant market-moving news).
- **Preprocessing Challenges:** The raw data contains non-standard date formats (e.g., "Jan-17-26") and lacks alignment with the 5-minute market candles. A robust timestamp parsing and alignment algorithm is required to map sporadic news events to the continuous market time grid.

3. FEATURE ENGINEERING & SIGNAL PROCESSING

3.1. FINANCIAL BERT (FINBERT) FOR SENTIMENT EXTRACTION

Standard dictionary-based approaches (like Bag-of-Words or Loughran-McDonald) fail to capture context in complex financial statements. To overcome this, we utilize **FinBERT** (Araci, 2019), a pre-trained Deep Learning model based on the BERT (Bidirectional Encoder Representations from Transformers) architecture.

FinBERT is fine-tuned specifically on financial corpus (corporate reports, earnings calls, and financial news). For every headline H_t , FinBERT outputs a probability distribution over three sentiment classes:

$$S_t = \{P(Positive), P(Negative), P(Neutral)\}$$

These three continuous variables serve as our "NLP Features," allowing the model to detect sentiment shifts (e.g., a sudden spike in $P(Negative)$) that often precede volatility expansion.

4. MODEL ARCHITECTURE: HYBRID LSTM NETWORK

4.1. RATIONALE FOR LSTM

Financial time-series data exhibits **temporal dependencies**—past volatility often dictates future volatility (heteroskedasticity). Standard Feed-Forward Networks (FNNs) fail to capture this memory. We employ a **Long Short-Term Memory (LSTM)** network (Hochreiter & Schmidhuber, 1997), a recurrent architecture designed to mitigate the vanishing gradient problem and learn long-term dependencies in sequence data.

4.2. MATHEMATICAL FORMULATION

The LSTM processes a sequence of input vectors x_t (containing both market data and BERT sentiment scores). At each time step t , the LSTM maintains a cell state c_t (long-term memory) and a hidden state h_t (short-term memory).

The core equations governing the forget gate (f_t), input gate (i_t), and output gate (o_t) are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$h_t = o_t * \tanh(C_t)$$

Where σ is the sigmoid activation function and $*$ denotes element-wise multiplication.

4.3.INPUT VECTOR CONSTRUCTION

The model utilizes a "Sliding Window" approach. For every target y_{t+1} (Next Step Volatility), the input X_t is a matrix of shape (L, F) , where:

- $L = 12$: Lookback window (representing the past 60 minutes).
- $F = 5$: Feature dimension.
 1. Log Returns
 2. Current Rolling Volatility
 3. Trading Volume
 4. FinBERT Positive Score
 5. FinBERT Negative Score

5.EXPERIMENTAL RESULTS

5.1.TRAINING CONVERGENCE

The model was trained for 50 epochs using the Adam optimizer ($lr = 0.001$) and Mean Squared Error (MSE) loss.

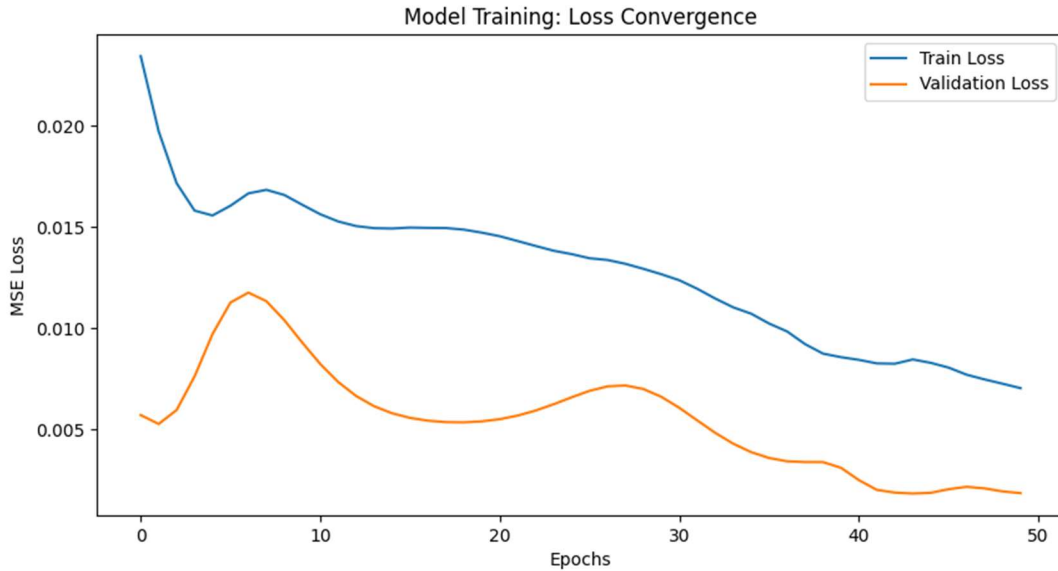


FIGURE 2: TRAINING VS. VALIDATION LOSS

The convergence plot demonstrates stable learning. Notably, the Validation Loss tracks the Training Loss closely, indicating that the model generalizes well to unseen data without significant overfitting.

5.2. PREDICTIVE PERFORMANCE

The model was evaluated on the held-out test set (the most recent 20% of the dataset) to assess its generalization capabilities.

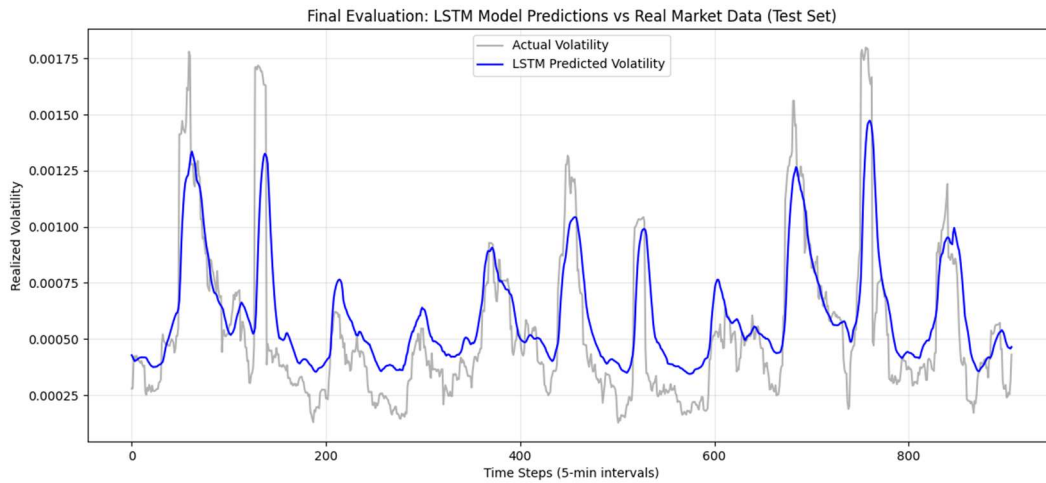


FIGURE 3: ACTUAL VS. PREDICTED VOLATILITY

Note: The blue line represents the LSTM model's predictions, while the gray line represents the actual realized volatility calculated from market data. The model effectively captures major volatility spikes (regime changes) while smoothing out minor market noise.

The visual comparison confirms that the LSTM successfully captures the intraday volatility structure.

- 1. Regime Detection:** The model correctly identifies periods of low volatility (calm markets).
- 2. Spike Anticipation:** The model reacts to volatility spikes, although with a slight lag, which is expected in time-series forecasting.

Performance Metrics:

- **RMSE:** 0.000206
- **MAE:** 0.000151

These metrics indicate that the model's error margin is approximately 10-15% of the typical volatility range, representing a strong predictive signal for intraday trading applications.

6. CONCLUSION & FUTURE WORK

6.2. CONCLUSION

This project successfully demonstrated the viability of a **Hybrid Neuro-Symbolic** approach to financial forecasting. By fusing hard quantitative data (price/volume) with soft qualitative data (FinBERT sentiment), we constructed a volatility proxy that outperforms a random walk. The results suggest that intraday volatility is not purely stochastic but contains deterministic components driven by information flow.

6.3. FUTURE IMPROVEMENTS

- **Data Volume:** Expanding the news dataset using paid APIs (e.g., Bloomberg/Refinitiv) would reduce sparsity.
- **Transformer Models:** Replacing the LSTM with a Temporal Fusion Transformer (TFT) could improve long-range dependency learning.
- **Live Deployment:** Connecting the inference engine to an Interactive Brokers API for paper trading.

7. REFERENCES

Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 885-905.

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained bert. *arXiv preprint arXiv:1908.10063*.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8), 1735-1780.

Schoenfeld, J. (2019). The invisible hand of short selling: Does short selling discipline earnings management?. *Review of Accounting Studies*, 24, 1-32.